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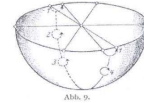
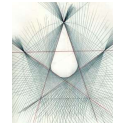
a.k.a. CSC 51074 EP

Geometric Modeling: Digital Representation and Analysis of Shapes



Today

- Practical Information
- Introduction to the course
- **First Lecture:**
 - Transformations, Homogeneous Coordinates
 - Representing Rotations, Euler Angles, Quaternions



Practical Information

- Instructors:



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TD1 (and pdf of slides for those who can't access moodle):
https://gitlab.inria.fr/geomerix/public/teaching/code-skeleton-csc_51074_ep_2026.git



Practical Information

Course website on Moodle (CSC_51074_EP 2026)

Let us know if you don't have access

- **9 Lectures and Practical Sessions (TD)**

Lectures: Tuesdays 13:30 – 15:30

TDs: Tuesdays 15:45 – 18:00
(PC rooms 35 & 36)

Lecture slides on Moodle (right) before each lecture

No Final Exam



Practical Information

Evaluation:

- **TDs: not graded** (blame AI)
 - mostly independent of each other
 - except for TD 4 (following ones will depend on it)
- **8 short quizzes: 16.6% each**
 - best 6 out of 8 grades
 - in effect: two wild cards, no excuses needed
 - BRING A PEN or PENCIL



Introduction to the course

What is **geometric modeling**?

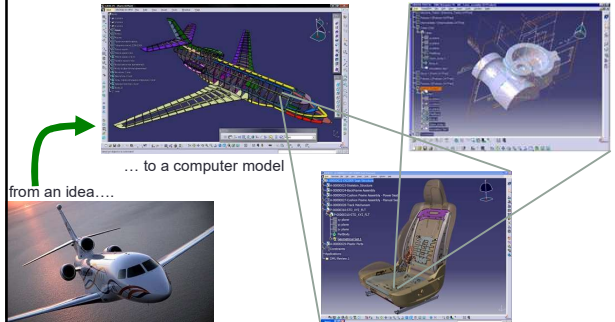
- Broad Goals:

To create mathematical models and practical tools for the digital representation & manipulation of (real or imagined) 2D/3D shapes



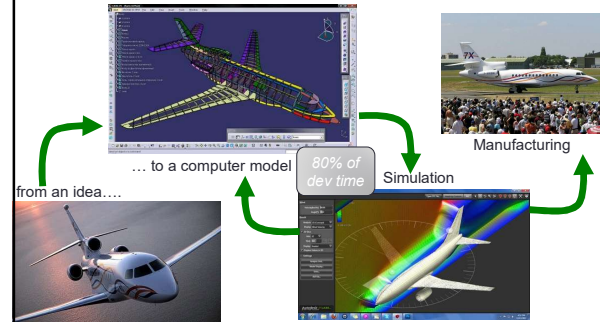
Why Geometric Modeling?

- Modern manufacturing pipeline: e.g. Dassault Falcon 7X



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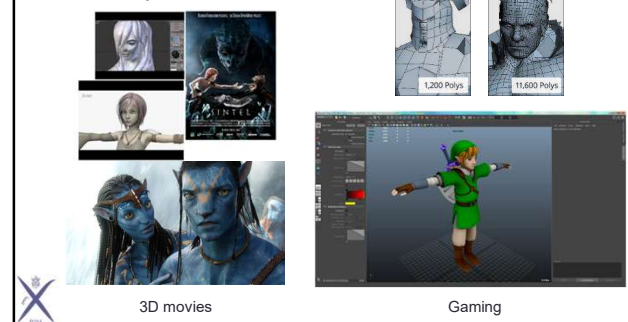
What is Digitally Modeled?

- Everything! (well, almost)



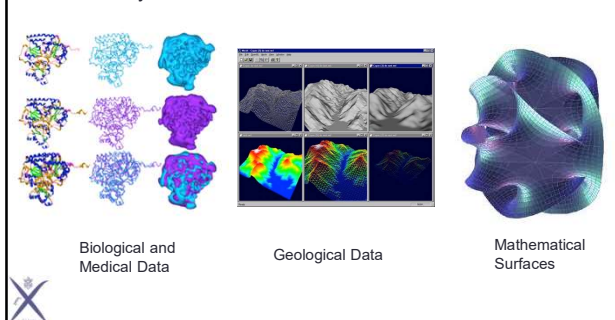
What is Digitally Modeled?

- And many more!



What is Digitally Modeled?

- And many more!



Why Geometric Modeling?

- A digital model allows easy **manipulation/visualization**
- Digital simulation is **much cheaper**
 - shape **optimization** (look, aerodynamics) done more easily
- **Comparison** across models is simpler
- Creation of **new models from existing ones** is easier

Why Geometric Modeling Now?



Konica Minolta Vivid 9i
~\$10K in 2004



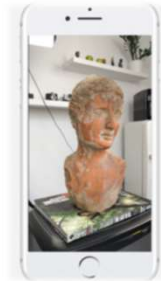
~\$150 in 2021

Why Geometric Modeling Now?

3D Scanning capabilities in recent devices



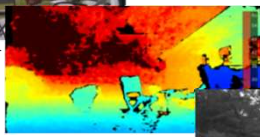
Sony Xperia XZ1



Why Geometric Modeling Now?



RGB color



Depth stream



Stereo infrared stream

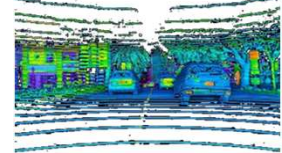
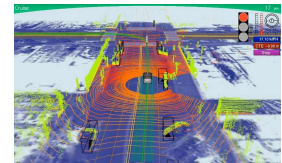


Intel REALSENSE

Why Geometric Modeling Now?



LIDAR sensors on self-driving cars



Why Geometric Modeling Now?



3D printers are becoming widely available



If you can model an object, you can create it at home

Why Geometric Modeling Now?

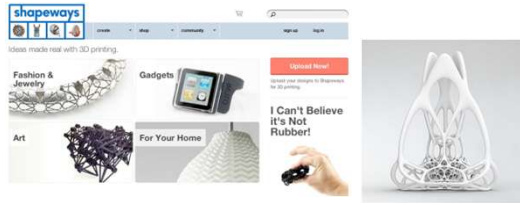


Personalized Prosthetics



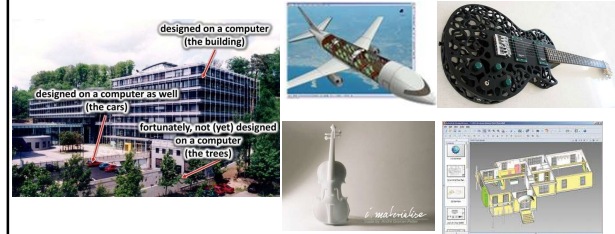
3D bioprinting of tissues and organs

Why Geometric Modeling Now?



Why Geometric Modeling Now?

Our ability to digitally model **directly affects the world we live in**

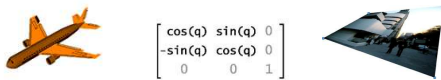


Current Grand Challenge: shape modeling & analysis still require expertise

What are we going to learn?

We proceed by increasing dimension:

- 0D: Points and their Transformations
 - Projective Geometry and Homogeneous Coordinates – **Today**



- 1D: Modeling Curves
 - recap of differential geometry
 - Parametric curves, polynomials, spline interpolation, subdivision



What are we going to learn?

We proceed by increasing dimension:

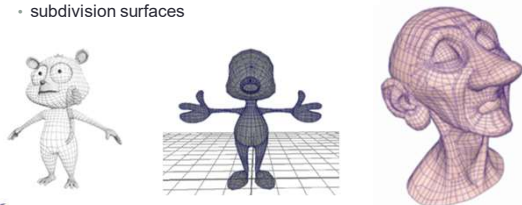
- 2D / 3D: Point Clouds & Implicit Surfaces
 - Analysis and cleanup – e.g. normal estimation
 - Reconstruction



What are we going to learn?

We proceed by increasing dimension and/or complexity

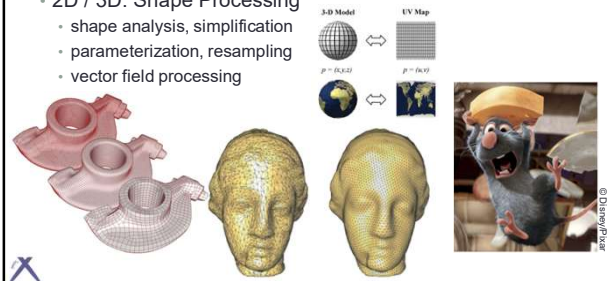
- 2D / 3D: Surface meshes
 - Bézier patches, triangle and polygonal meshes
 - subdivision surfaces



What are we going to learn?

We proceed by increasing dimension:

- 2D / 3D: Shape Processing
 - shape analysis, simplification
 - parameterization, resampling
 - vector field processing



What are we going to learn?

We proceed by increasing dimension:

- 2D / 3D: Shape Processing
 - Shape Comparison / Deformation / Search + data-driven methods



Questions?

